

3(a)(i)	<i>any 4 of:</i> 1 chromosomes become visible ; 2 condensation / coiling / supercoiling ; A description 3 nuclear envelope or nuclear membrane, disintegrates / disappears ; 4 nucleolus, disintegrates / disappears ; 5 centrioles migrate to (opposite) poles ; I centrosomes 6 spindle forms / microtubules assemble ; 7 synapsis / bivalents form / homologous chromosomes pair up ; 8 chiasmata formation / crossing over ;	4
3(a)(ii)	<i>any 2 of:</i> 1 (homologous chromosomes / bivalents) line up randomly on the, equator / metaphase plate ; 2 produces (many) different chromosome combinations ; 3 number of possible combinations is 2^n ; 4 new combinations of maternal and paternal alleles ;	2
3(b)(i)	<i>any 2 of:</i> 1 protoctist uses up nutrients ; 2 loss of appetite / inability to feed ; 3 <i>idea of</i> impaired absorption / diarrhoea ; 4 (so) less nutrients available for, growth / gain of body mass / increase in biomass ; 5 AVP ; e.g. <i>idea of</i> energy used to fight infection / <i>ref. to</i> immune response	2

Question	Answer	Marks
3(b)(ii)	(chicken) body temperature / competition with microbes / presence of drugs / presence of plant extract ;	1
3(c)(i)	<u>t-test</u> ;	1
3(c)(ii)	<i>any 3 of:</i> 1 two labelled bell-shaped curves ; 2 high dose curve shifted to left of low dose curve ; 3 high dose curve peak to left of 2063 ; 4 high dose curve, narrower / peak higher ;	3

Question	Answer	Marks
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